

# SWORD v17c Beta Release Notes

**Version:** v17c beta 0.0.4 **Date:** March 2026 **Authors:** James H. Gearon, Tamlin M. Pavelsky, Niek Collot d'Escury **Base version:** SWORD v17b (March 2025, UNC)

## Changelog

### 0.0.4 (March 2026)

- **Node propagation to 11.1M nodes.** Five v17c columns (`best_headwater`, `best_outlet`, `pathlen_hw`, `pathlen_out`, `subnetwork_id`) were NULL on all nodes. Now propagated from parent reaches.
- **Dijkstra ghost outlet fix.** Ghost reaches (`type=6`) with `out_degree=0` no longer report `dist_out_dijkstra=0`. All sinks are used as Dijkstra sources for full coverage (94–99% per region); ghost sinks receive NULL. Real outlet counts: NA=7, SA=1, EU=2, AF=1, AS=11, OC=1.
- **Bifurcation routing fix (V023).** At multi-successor nodes, `rch_id_dn_main` now uses score-based ranking instead of the mainstem chain, which was constrained to within-group successors. V023 violations:  $2 \rightarrow 0$ .
- **hydro\_dist\_hw computed.** Mainstem distance from headwater via `rch_id_up_main` chain walk (mirror of `hydro_dist_out`). Was stale from a prior pipeline run.
- **facc monotonicity fix (T003).** 408 reaches corrected via iterative downstream propagation (4 passes). T003 violations:  $392 \rightarrow 5$ .
- **Routing weights retrained on effective\_slope.** SWOT `slope_obs_p50` (where reliable and  $n \geq 5$ ) replaces MERIT DEM slope in routing score training. Slope share:  $8\% \rightarrow 3\%$ , width:  $67\% \rightarrow 71\%$ . CV accuracy unchanged (88.5%).
- **Variable reference updated.** 7 missing variables added, 8 type mismatches fixed, `cl_ids` shape corrected to `cl_id_min/cl_id_max`.

### 0.0.3 (March 2026)

- **Routing weights learned from human labels.** Replaced the handcrafted lexicographic 3-tuple (`effective_width`, `log_facc`, `pathlen`) with a weighted scalar score trained on 1,967 human-labeled junction decisions:  $1.97 \log_{10}(ew) + 0.23 \log_{10}(facc) - 0.23 \log_{10}(slope) + 0.23 \log_{10}(pathlen) - 0.29 * stream\_order$ . Two new signals vs prior releases: slope (negative = prefer lower gradient) and `stream_order`. All routing functions use the same score to prevent divergence. (Weights retrained in 0.0.4; see above.)
- **Caroline's reviewer fixes synced.** 127 C001 lakeflag fixes (NA) and 10 C004 type fixes (NA) from the Streamlit reviewer app.
- **NA PostgreSQL geometry fix.** 38,696 NA reaches had NULL geometry in PostgreSQL — v17b geometry overwrite via dblink silently failed. Rewrote to read from v17b GPKG files on disk with verification gates.
- Fixed `dn_node_id`, `up_node_id`, and `node_order` for 810 flow-corrected reaches where node ordering was stale (based on v17b `dist_out`). 639 reaches now have `dn_node_id != min(node_id)`, reflecting the inverted flow direction.
- Removed redundant `rch_id_up_1..4` / `rch_id_dn_1..4` vector variables from NetCDF export. Only the  $[4, N]$  matrices `rch_id_up` and `rch_id_dn` are exported, matching v17b format.
- Fixed `swot_orbits` NetCDF type from string back to int64 (matching v17b).
- Fixed 15 integer columns widened from int32 to int64 in NetCDF export (`n_nodes`, `lakeflag`, `n_rch_up`, `stream_order`, etc.). All shared variable types now match v17b exactly.
- Synced `rch_id_up_1..4` / `rch_id_dn_1..4` DB columns from `reach_topology` table (694 were stale after flow corrections).

### 0.0.2 (March 2026)

- Renamed `is_mainstem_edge` to `is_mainstem`

- Mainstem algorithm refactored: computed per `main_path_id` group (see Section 2.1)
- Fixed `n_rch_up/n_rch_down` stale counts at 148 flow-corrected reaches
- Recomputed `facc` at 807 flow-corrected reaches using topological propagation; resolves all F006 junction conservation violations and T003 monotonicity violations at these reaches
- OC reach 51111300061 incomplete split fully reverted to v17b state (434 orphan centerlines, 73 orphan nodes restored)

### 0.0.1 (March 2026)

- Initial v17c beta release
- Includes `node_order`, `dn_node_id`, `up_node_id` (node ordering within reaches)

## 1. Overview

SWORD v17c extends v17b with three additions: computed mainstem topology, SWOT observation statistics, and flow accumulation corrections. No reaches, nodes, or centerlines were added or removed. v17c contains the same 248,673 reaches, 11.1M nodes, and 66.9M centerline points as v17b across all six regions (NA, SA, EU, AF, AS, OC).

Each region is distributed as a single NetCDF4 file (`{region}_sword_v17c_beta.nc`). The group structure matches v17b (centerlines, nodes, reaches), and the `area_fits` and `discharge_models` subgroups under reaches pass through from v17b unchanged. Reach and node ordering within each file matches v17b.

Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) match v17b values across all formats (NetCDF, DuckDB, PostgreSQL).

All new variables use a fill value of -9999 where no observation or computation produced a value.

For a complete variable catalog, see `v17c_variable_reference.md`.

## 2. New Variables

### 2.1 Mainstem Topology (reaches group)

`is_mainstem` is computed per `main_path_id` group: each group (a mainstem path plus its tributary branches) gets one canonical chain, identified by a greedy walk from the group's shared `best_headwater`. At each junction the algorithm selects the upstream branch with the highest weighted routing score:  $2.02 \cdot \log_{10}(ew) + 0.17 \cdot \log_{10}(facc) - 0.08 \cdot \log_{10}(slope) + 0.35 \cdot \log_{10}(pathlen) + 0.23 \cdot stream\_order$ . These weights were learned from 1,967 human-labeled junction decisions via logistic regression on pairwise  $\log_{10}$ -difference features, using SWOT slope where reliable (else MERIT DEM). The negative slope weight captures the geomorphic pattern that mainstem channels have lower gradients than tributaries. Mainstem reaches within each group are then assigned `rch_id_up_main` / `rch_id_dn_main` from the chain; non-mainstem reaches use the same weighted score. Ghost reaches (`type=6`) are excluded from mainstem but still participate in routing topology. ~10% of mainstem reaches have no mainstem neighbor at `main_path_id` group boundaries — this is expected by design.

| Variable                       | Type    | Units  | Description                                                                          |
|--------------------------------|---------|--------|--------------------------------------------------------------------------------------|
| <code>dist_out_dijkstra</code> | float64 | meters | Dijkstra shortest-path distance to any network outlet                                |
| <code>hydro_dist_out</code>    | float64 | meters | Mainstem distance to <code>best_outlet</code> via <code>rch_id_dn_main</code> chain  |
| <code>hydro_dist_hw</code>     | float64 | meters | Distance from <code>best_headwater</code> via <code>rch_id_up_main</code> chain walk |

| Variable                    | Type    | Units  | Description                                                                         |
|-----------------------------|---------|--------|-------------------------------------------------------------------------------------|
| <code>rch_id_up_main</code> | int64   | —      | Main upstream neighbor <code>reach_id</code> (mainstem-preferred)                   |
| <code>rch_id_dn_main</code> | int64   | —      | Main downstream neighbor <code>reach_id</code> (mainstem-preferred)                 |
| <code>best_headwater</code> | int64   | —      | Routing-score-prioritized headwater <code>reach_id</code> for the network component |
| <code>best_outlet</code>    | int64   | —      | Routing-score-prioritized outlet <code>reach_id</code> for the network component    |
| <code>pathlen_hw</code>     | float64 | meters | Cumulative path length from <b><code>best_headwater</code></b>                      |
| <code>pathlen_out</code>    | float64 | meters | Cumulative path length to <b><code>best_outlet</code></b>                           |
| <code>is_mainstem</code>    | int8    | —      | 1 if reach is on a mainstem path, 0 otherwise                                       |
| <code>main_path_id</code>   | int64   | —      | Unique identifier for each mainstem path group                                      |
| <code>subnetwork_id</code>  | int32   | —      | Connected component ID (Pfaffstetter-offset, globally unique; see Section 4)        |
| <code>dn_node_id</code>     | int64   | —      | Node ID at the downstream end of the reach (lowest <code>dist_out</code> )          |
| <code>up_node_id</code>     | int64   | —      | Node ID at the upstream end of the reach (highest <code>dist_out</code> )           |

Five of these variables also appear at node level: `subnetwork_id`, `best_headwater`, `best_outlet`, `pathlen_hw`, and `pathlen_out`.

`node_order` is a node-level variable (not in the reaches table): 1-based position within a reach, ordered by `dist_out` ascending (1 = downstream end, n = upstream end).

## 2.2 SWOT Observation Statistics

Percentile-based summaries computed from available SWOT observations. All percentile, range, and MAD variables share the units of the underlying measurement.

### Reaches and nodes:

| Variable                     | Type    | Units  | Description                       |
|------------------------------|---------|--------|-----------------------------------|
| <code>wse_obs_p10–</code>    | float64 | meters | WSE percentiles (10th through     |
| <code>wse_obs_p90</code>     |         |        | 90th, in steps of 10)             |
| <code>wse_obs_range</code>   | float64 | meters | WSE observation range (p90 - p10) |
| <code>wse_obs_mad</code>     | float64 | meters | WSE median absolute deviation     |
| <code>width_obs_p10–</code>  | float64 | meters | Width percentiles                 |
| <code>width_obs_p90</code>   |         |        |                                   |
| <code>width_obs_range</code> | float64 | meters | Width observation range           |
| <code>width_obs_mad</code>   | float64 | meters | Width median absolute deviation   |
| <code>n_obs</code>           | int32   | —      | Total SWOT observation count      |

### Reaches only:

| Variable                        | Type    | Units | Description                                   |
|---------------------------------|---------|-------|-----------------------------------------------|
| slope_obs_p10–<br>slope_obs_p90 | float64 | m/km  | Slope percentiles                             |
| slope_obs_range                 | float64 | m/km  | Slope observation range                       |
| slope_obs_mad                   | float64 | m/km  | Slope median absolute deviation               |
| slope_obs_adj                   | float64 | m/km  | Adjusted slope                                |
| slope_obs_slopeF                | float64 | —     | Slope F-statistic                             |
| slope_obs_reliable              | int32   | —     | 0 = unreliable, 1 = reliable                  |
| slope_obs_quality               | int32   | —     | Integer quality category (0–8; see Section 3) |
| slope_obs_n                     | int32   | —     | Number of slope observations                  |
| slope_obs_n_passes              | int32   | —     | Number of SWOT passes used                    |
| slope_obs_q                     | int32   | —     | Bitfield quality flag (see Section 3)         |

## 2.3 Flow Accumulation Corrections

A two-stage denoise pipeline corrected flow accumulation (**facc**) values to address three systematic error modes in MERIT Hydro’s D8 (eight-direction flow routing) upstream area: bifurcation cloning, junction inflation, and raster-vector misalignment. The pipeline corrected 95,913 of 248,673 reaches (38.6%). Uncorrected reaches retain v17b values. See `facc_correction_methodology.md` for the full algorithm description.

| Variable            | Type    | Group          | Description                                                                                    |
|---------------------|---------|----------------|------------------------------------------------------------------------------------------------|
| <b>facc</b>         | float64 | reaches, nodes | Flow accumulation ( $\text{km}^2$ ). Corrected values where applicable; v17b values otherwise. |
| <b>facc_quality</b> | int32   | reaches, nodes | 1 = corrected by denoise_v3; fill_value = not flagged                                          |

After correction, junction conservation violations (downstream `facc` < sum of upstream `facc`) are resolved in all regions. In 0.0.2, `facc` was additionally recomputed at 807 flow-corrected reaches via topological propagation, resolving remaining monotonicity violations at those reaches.

## 2.4 Other New or Updated Variables

| Variable          | Type   | Group   | Description                                                                                                                                                                                     |
|-------------------|--------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>type</b>       | int32  | reaches | Reach classification (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not present in v17b NetCDF; added in v17c so NetCDF users can filter by reach type without needing the database. |
| <b>dl_grod_id</b> | int64  | reaches | DL-GROD (Deep Learning Global River Obstruction Database; He et al. 2025) dam/obstruction ID                                                                                                    |
| <b>edit_flag</b>  | string | reaches | Tag for manually edited reaches (e.g., <code>lake_sandwich</code> , <code>harp_lake</code> )                                                                                                    |

### 3. Flag Encoding Reference

#### facc\_quality

| Value        | Meaning                                                            |
|--------------|--------------------------------------------------------------------|
| 1            | <code>denoise_v3</code> — corrected by the facc denoising pipeline |
| -9999 (fill) | Not flagged; facc unchanged from v17b                              |

CF attributes: `flag_values = [1]`, `flag_meanings = "denoise_v3"`.

#### slope\_obs\_quality

| Value | Meaning                            |
|-------|------------------------------------|
| 0     | reliable                           |
| 1     | <code>small_negative</code>        |
| 2     | <code>moderate_negative</code>     |
| 3     | <code>large_negative</code>        |
| 4     | negative                           |
| 5     | <code>below_ref_uncertainty</code> |
| 6     | <code>high_uncertainty</code>      |
| 7     | <code>noise_high_nobs</code>       |
| 8     | <code>flat_water_noise</code>      |

CF attributes: `flag_values = [0,1,2,3,4,5,6,7,8]`, `flag_meanings = "reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"`.

#### slope\_obs\_reliable

| Value        | Meaning                                   |
|--------------|-------------------------------------------|
| 0            | Unreliable                                |
| 1            | Reliable                                  |
| -9999 (fill) | Not computed (no SWOT slope observations) |

#### slope\_obs\_q (bitfield)

| Bit | Value | Meaning              |
|-----|-------|----------------------|
| 1   | 1     | Negative slope       |
| 2   | 2     | Low number of passes |
| 3   | 4     | High variance        |
| 4   | 8     | Extreme value        |
| 5   | 16    | Clipped              |

Flags combine by addition. A value of 0 indicates no quality issues. Example: 5 = negative slope (1) + high variance (4).

is\_mainstem

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| Value | Meaning          |
|-------|------------------|
| 0     | Not on mainstem  |
| 1     | On mainstem path |

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#### 4. Known Limitations

- **SWOT observation coverage:** SWOT statistics are fill\_value (-9999) for reaches and nodes lacking SWOT data.
- **facc correction scope:** 95,913 reaches corrected (38.6%); the remaining 152,761 retain v17b values. Node-level facc propagates from the parent reach.
- **Lake sandwich corrections:** 1,252 reaches reclassified to lakeflag = 1 where a narrow, shorter-than-neighbor reach sat between lake reaches (tagged edit\_flag = "lake\_sandwich"). ~1,755 similar cases remain (narrow connecting channels, chains).
- **HarP lake corrections:** 7,425 reaches reclassified from lakeflag = 0 (river) to lakeflag = 1 (lake) based on HarP v1.1 (Hydrography and River Planform) lake classification data. 200,201 child nodes updated to match. Tagged edit\_flag = "harp\_lake". Existing tags preserved (comma-delimited when multiple apply).
- **area\_fits and discharge\_models:** Direct copies from v17b. Not recomputed against v17c facc or SWOT values.
- **subnetwork\_id vs network:** subnetwork\_id uses Pfafstetter- offset enumeration (globally unique). v17b network uses per-region 1-based IDs. Different component counts (v17c finds more via weakly connected components; 19 subnetworks span multiple v17b networks). network is retained unchanged from v17b.
- **Topology reciprocity gaps (resolved):** 151 non-reciprocal pairs (both reaches listing each other as upstream) were introduced during flow correction revert. Fixed by completing the revert for all affected reciprocal entries. Zero non-reciprocal pairs remain.
- **Flow correction oscillation:** 389 reaches (0.16%) in AF/AS/EU/NA/SA had ambiguous WSE slope signals causing bidirectional flow correction scores. These were reverted to v17b topology.
- **main\_path\_id consistency:** 3,134 reaches have main\_path\_id values inconsistent with current (best\_headwater, best\_outlet) tuples (V013- V015 lint checks). 80 reaches in NA have best\_headwater pointing to non-headwater reaches. Requires recomputing main\_path\_id from current headwater/outlet assignments.
- **River naming:** 51.2% of reaches are unnamed (NODATA), ranging from 26% (AF) to 69% (OC). 2.6% of mainstem 1:1 links have local name discontinuities (name changes between adjacent reaches with no junction).
- **Width fill values (A003):** ~1,266 reaches have width=0 (unmeasured) or width=-1 (GRWL lake fill). These are v17b fill values, not data errors. Present in both v17b and v17c unchanged. The A003 lint check is downgraded to WARNING for this reason.
- **lakeflag/type mismatch:** ~5,770 reaches have lakeflag=1 (lake) but type=1 (river), introduced by HarP and lake-sandwich corrections which updated lakeflag but not type. type is encoded in the last digit of reach\_id, so changing it would change reach IDs. Policy for reach\_id changes under discussion. Affects users filtering on both lakeflag and type at river-lake boundaries. Tracked in issue #208.

## 5. Quality Audits

Validation checks performed on the v17c data:

| Audit                                            | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Geometry</b>                                  | DuckDB geometries (rebuilt from NetCDF) lack endpoint overlap vertices present in v17b (210,533 reaches affected: 173K +1 point, 37K +2 points). <code>reach_length</code> unchanged. Reach coordinate columns ( <code>x</code> , <code>y</code> , <code>x_min</code> , <code>x_max</code> , <code>y_min</code> , <code>y_max</code> ) copied from v17b to ensure consistency across all formats.                                                                                                           |
| <b>n_nodes / reach_length<br/>path_freq gaps</b> | Internally consistent. Zero N008/G002/G003 violations. v17b had 4,952 connected non-ghost reaches with invalid <code>path_freq</code> (0 or -9999). Resolved in v17c; remaining nodata values are correctly attributed to ghost reaches (type=6).                                                                                                                                                                                                                                                           |
| <b>subnetwork_id</b>                             | 3,027 components across 248,673 reaches verified. Pfafstetter banding correct. Zero cross-region collisions. 19 subnetworks (0.6%) span multiple v17b networks (expected).                                                                                                                                                                                                                                                                                                                                  |
| <b>Topology integrity</b>                        | T001 ( <code>dist_out_dijkstra</code> monotonicity), T012 (referential integrity), T013 (self-reference), T014 (bidirectional): all pass. T005/T007: zero non-reciprocal edges (151 from incomplete flow correction revert resolved in beta 0.0.1). Note: v17b <code>dist_out</code> is stale at 807 flow-corrected reaches where topology direction was updated but <code>dist_out</code> retains its v17b value — use <code>dist_out_dijkstra</code> or <code>hydro_dist_out</code> for distance routing. |
| <b>n_rch_up/n_rch_down</b>                       | 148 scalar count mismatches corrected (flow corrections flipped <code>reach_topology</code> but did not recalculate counts). Zero mismatches across all 248,673 reaches.                                                                                                                                                                                                                                                                                                                                    |
| <b>OC reach split revert</b>                     | Incomplete <code>break_reaches()</code> split of OC reach 51111300061 (434 orphan centerlines, 73 orphan nodes) fully reverted to v17b state.                                                                                                                                                                                                                                                                                                                                                               |
| <b>River name formatting</b>                     | 291 formatting issues corrected (separators, whitespace). Automated checks now enforce “;” separator and alphabetical ordering.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Flow direction</b>                            | 1,112 experimental topology flips reverted after causing 30K disconnected reaches. 807 reaches across all regions retain corrected topology that differs from v17b (175 sections, median 5 reaches/section). OC has 119 of these (26 SWOT-validated sections). Non-OC corrections were retained from the flow correction pipeline; <code>dist_out</code> is stale at these reaches — use <code>dist_out_dijkstra</code> .                                                                                   |
| <b>HarP lake corrections</b>                     | 7,425 reaches reclassified lakeflag 0 to 1 from HarP v1.1 data. 200,201 nodes propagated. Tagged <code>edit_flag</code> = “harp_lake”.                                                                                                                                                                                                                                                                                                                                                                      |

For POM (Pierre-Olivier Malaterre) validation results, see `pom_validation_report.md`.

## 6. File Format

- **Format:** NetCDF4 (one file per region)
  - **Naming:** {region}\_sword\_v17c\_beta.nc where region is na, sa, eu, af, as, oc
  - **Groups:** centerlines, nodes, reaches
    - reaches/area\_fits and reaches/discharge\_models subgroups (from v17b)
  - **Ordering:** Reach, node, and centerline arrays match v17b ordering
  - **Fill value:** -9999 for all numeric variables (int32, int64, float64)
  - **Checksums:** SHA256 hashes listed in SHA256SUMS\_{version}.txt
  - **Additional formats:** GeoPackage and GeoParquet exports available (reaches and nodes per region, with geometry)
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## 7. Methodology Documentation

| Document                              | Description                                                  |
|---------------------------------------|--------------------------------------------------------------|
| facc_correction_methodology.md        | Facc denoise algorithm, detection rules, correction model    |
| pom_requests_summary.md               | POM validation check tracker (19 checks, production results) |
| pom_validation_report.md              | POM validation production results                            |
| v17c_variable_reference.md            | Complete variable catalog for NetCDF export                  |
| SWORD_v17b_Technical_Documentation.md | v17b baseline reference                                      |